

The Cartographic Theory of the Standard Nuclear oliGARCHy:

Globe, Projection, Satellite Observation, and Map-Making
A Reciprocal Technology-Transfer Framework between Portuguese Cartography and
SNoG Earth Observation

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

A globe and a map are not the same object. A globe represents a curved terrestrial surface directly, whereas a planar map is produced through a cartographic transformation whose distortion is unavoidable and whose acceptable compromises depend on purpose [1, 2, 4]. This paper develops a compact interdisciplinary framework for a reciprocal technology-transfer arrangement between Portuguese cartography and the Earth-observation capabilities posited within the Standard Nuclear oliGARCHy (SNoG). The central distinction is between transferring products, such as individual maps, and transferring the productive technology that generates maps from observed spatial states.

The paper formalizes the cartographic pipeline as

$$X_t \longrightarrow Y_t \longrightarrow \hat{X}_t \longrightarrow P_{\theta_t}(\hat{X}_t) \longrightarrow M_t, \quad (1)$$

where X_t is the terrestrial state, Y_t is remote-sensing observation, $\hat{X}_t$ is an estimated state, P_{θ_t} is a selected map projection, and M_t is the resulting map. The economic argument is expressed as an option-value result: when the recipient can select among projection technologies after learning the state of the world, technology transfer can weakly dominate transfer of a fixed map and can strictly dominate under genuine state-contingent uncertainty. The analysis connects cartography to real options, asset pricing, econometrics, physics, information theory, chemistry, biology, psychology, political science, diplomacy, international relations, welfare economics, warfare economics, computer science, data science, and artificial intelligence. Existing SNoG papers on political institutions, nuclear diplomacy, international relations, inductance, and commerce provide the institutional and economic background [15, 16, 17, 18, 14].

Lopo Homem is used here as a historical representative of the Portuguese cartographic tradition rather than as a claimed present-day author. Raghuram G. Rajan provides a natural financial-economics perspective on technology, institutions, and optionality; his current institutional affiliation is Chicago Booth [6]. The paper's central policy proposal is therefore reciprocity: Portugal need not merely receive SNoG-generated maps, and SNoG need not merely receive inherited Portuguese maps. Both sides should exchange the technologies that generate, interpret, validate, and adapt spatial representations.

The paper ends with "The End"

1 Introduction

Cartography begins with a deceptively simple distinction: the Earth is a curved surface, while most practical maps are planar representations. A map projection is consequently a mathematical transformation from a curved surface, usually modeled by a sphere or ellipsoid, to a plane. Such a transformation necessarily introduces distortion; there is no single projection that is universally best for all purposes [1, 2, 4].

This fact is not merely a technical footnote. It changes the economics of geographic information. A finished map is an output conditional on an underlying spatial state and a chosen representation rule. Cartographic technology is the larger capability that permits the producer to observe, transform, select, update, compare, and generate maps for new states of the world.

The historical Portuguese example is especially useful. The Miller Atlas of 1519 was produced collaboratively by Pedro Reinel, Jorge Reinel, and Lopo Homem, and the Library of Congress preserves digitized materials documenting this cartographic work [5]. The relevant lesson is not antiquarian. It is institutional: navigational knowledge was embedded in a technological system for representing maritime space.

The SNoG framework introduces a complementary technological perspective. Earlier SNoG work treats diplomacy, institutions, and strategic interaction as coupled systems, and its inductance framework asks whether a new participant contributes positive systemic value rather than merely occupying another slot [15, 16, 17, 18]. The present paper applies that logic to cartography and Earth observation.

The proposed exchange is therefore:

$$\text{Portuguese Cartographic Technology} \longleftrightarrow \text{SNoG Earth-Observation Technology}. \quad (2)$$

The claim is not that one side lacks the other. Portugal possesses a deep cartographic tradition; the SNoG architecture is here modeled as possessing modern observation systems capable of generating geographic information. The economic objective is to make the two technological stacks interoperable.

2 Globe, Map, and Projection

Let the terrestrial globe be a manifold $\mathcal{G}$. For a spherical benchmark with radius R , the local metric can be written as

$$ds^2 = R^2 (d\varphi^2 + \cos^2 \varphi d\lambda^2). \quad (3)$$

A map projection is a transformation

$$P_\theta : \mathcal{G} \rightarrow \mathbb{R}^2, \quad (4)$$

where θ denotes the choice of projection family and parameters.

If J_θ denotes the Jacobian of the projection in local coordinates, the planar metric induced by the transformation is represented locally by

$$G_\theta = J_\theta^\top J_\theta. \quad (5)$$

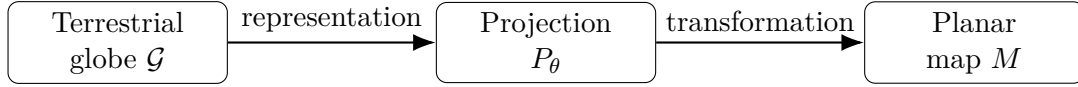
Distortion arises because the planar metric generally cannot equal the spherical metric everywhere. The geometric reason is captured by Gauss’s Theorema Egregium: an isometry cannot map a surface of positive Gaussian curvature globally onto a plane of zero Gaussian curvature. The practical cartographic consequence is the familiar trade-off among area, angle, distance, and direction [2, 1, 4].

2.1 A Minimal Distortion Theorem

Theorem 1 (No lossless global planar projection). There is no globally distance- and angle-preserving mapping of a sphere onto a plane.

Proof. The sphere has positive Gaussian curvature $1/R^2$, whereas the plane has zero Gaussian curvature. An isometry preserves intrinsic Gaussian curvature. Hence a global isometry from the sphere to the plane is impossible. $\square$

This theorem is established by classical differential geometry; the present paper uses it only as the structural foundation for the economic distinction between a globe and a map [2, 4].



Distortion is unavoidable; its allocation is purpose-dependent.

Figure 1: The globe–projection–map distinction.

The USGS likewise emphasizes that there is no universally best projection: the mapmaker chooses a representation suited to the task, while specialized projections can be designed for particular applications, including satellite mapping [1, 2].

3 Observation Before Representation

A modern cartographic system contains an additional layer that did not exist in the same form in early nautical charting: remote observation. Let the latent terrestrial state be X_t , including geographic, environmental, infrastructural, demographic, and logistical variables. Let the sensor system produce observations Y_t . A generic state-space representation is

$$Y_t = H_t(X_t) + \varepsilon_t, \quad (6)$$

with observation noise ε_t .

The estimated state is therefore

$$\hat{X}_t = \mathbb{E}[X_t \mid \mathcal{F}_t], \quad (7)$$

where $\mathcal{F}_t$ is the information set. This connects cartography directly to filtering, statistical inference, and data assimilation [7].

Map production then becomes

$$M_t = P_{\theta_t}(\hat{X}_t). \quad (8)$$

This decomposition is economically consequential. The map is only one realization of a much larger information pipeline:

$$\text{Sensing} \rightarrow \text{Inference} \rightarrow \text{Projection} \rightarrow \text{Representation} \rightarrow \text{Decision}. \quad (9)$$

Satellite observation therefore complements rather than replaces cartography. The USGS history of satellite-oriented projection design, including the Space Oblique Mercator, is a direct illustration of the interaction between remote sensing and projection theory [3].

4 Technology versus Product

The principal economic distinction of this paper is between a fixed product and a productive technology.

Let Θ denote a set of available cartographic technologies, projections, or projection configurations. A fixed product uses one element $\theta_0 \in \Theta$. A technology transfer enables the recipient to choose among elements of Θ .

Suppose the world state is $s \in \Omega$, the value of the resulting representation is $u(m, s)$, and use of technology θ has cost $c(\theta)$. Product transfer has value

$$V_{\text{product}} = \max_{\theta \in \Theta} \mathbb{E} [u(P_{\theta}(X), s) - c(\theta)]. \quad (10)$$

Technology transfer creates the possibility of state-contingent choice after information arrives:

$$V_{\text{technology}} = \mathbb{E} \left[\max_{\theta \in \Theta} \{u(P_{\theta}(X), s) - c(\theta)\} \right]. \quad (11)$$

4.1 Option-Value Proposition

Proposition 1 (Technology-transfer dominance). Under the information structure above,

$$V_{\text{technology}} \geq V_{\text{product}}. \quad (12)$$

The inequality is strict whenever the optimal choice of θ varies across states with positive probability and the relevant value differences are nonzero.

Proof. For every state s ,

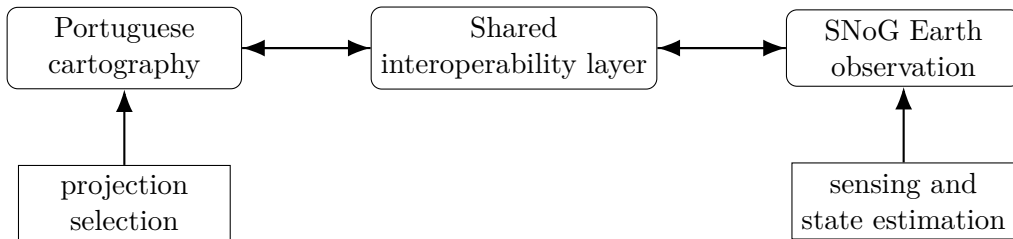
$$\max_{\theta \in \Theta} z_{\theta}(s) \geq z_{\theta_0}(s) \quad (13)$$

for every fixed θ_0 . Taking expectations gives the weak inequality. If different θ 's are optimal on distinct positive-probability subsets of Ω , no single fixed θ_0 achieves the state-by-state maximum everywhere, so the inequality is strict. $\square$

This is the core real-options argument. The recipient is not merely receiving today's map. The recipient receives an option to generate future maps conditional on future states of the world. The logic is consistent with the option-based orientation already present in the SNoG international-relations framework, where future strategic states alter the value of present commitments [17].

5 The Reciprocal Technology-Transfer Architecture

The proposed arrangement is reciprocal. Portugal contributes cartographic methods, design knowledge, projection expertise, historical geographic knowledge, and representational judgment. The SNoG side contributes modern Earth-observation capabilities, computational processing, state estimation, and automated map generation.



New observations + new projections $\rightarrow$ new maps

Figure 2: Reciprocal technology transfer rather than one-way product transfer.

The appropriate transaction is consequently not

$$\text{SNoG maps} \longrightarrow \text{Portugal}, \quad (14)$$

but

$$\text{Portuguese cartography} \longleftrightarrow \text{SNoG Earth observation and computation.} \quad (15)$$

This arrangement preserves the technological autonomy of both sides while expanding the joint feasible set.

6 Financial Economics and Optionality

Raghuram G. Rajan is an especially natural conceptual financial economist for this problem because the central question is not simply output maximization but institutional design under uncertainty, incentives, and future opportunities. Rajan is the Katherine Dusak Miller Distinguished Service Professor of Finance at the University of Chicago Booth School of Business and has research interests including banking, corporate finance, economic development, and political economy [6].

The technology-transfer principle can be interpreted as a real option. A map is a current payoff. A cartographic technology is a capability to choose future payoffs after information arrives.

Let F_t denote the filtration available at time t , and let $V_t(\theta)$ denote the value of projection technology θ . The value of flexibility can be summarized by

$$\Omega_t = \mathbb{E}_t \left[\max_{\theta \in \Theta} V_{t+1}(\theta) \right] - \max_{\theta \in \Theta} \mathbb{E}_t [V_{t+1}(\theta)]. \quad (16)$$

By Proposition 1,

$$\Omega_t \geq 0. \quad (17)$$

This is not an argument that technology transfer always has higher contractual value. Technology can be expensive to learn, maintain, govern, or secure. The correct comparison is therefore

$$\Omega_t - C_{\text{transfer}} - C_{\text{training}} > 0. \quad (18)$$

The commercial theory developed elsewhere in the SNoG corpus similarly distinguishes continuation value from stopping value and treats commerce as a stochastic process containing return, persistence, loss, insurance, and sequential choice [14]. The cartographic case is another instance of the same deeper distinction between a realized output and a capability to generate future outputs.

7 Econometrics, Statistics, and Data Science

The transfer arrangement is empirically testable. Let Y_i be an outcome variable measuring map quality, decision quality, or operational usefulness for application i . Let T_i indicate technology access, P_i indicate product-only access, and X_i denote controls. A basic model is

$$Y_i = \alpha + \beta T_i + \gamma P_i + X_i^T \delta + \varepsilon_i. \quad (19)$$

A stronger design would exploit staged adoption, randomized task assignment, difference-in-differences, or another credible identification strategy [11, 12]. The dependent variable should not be restricted to visual quality. Possible measures include positional accuracy, route-error reduction, decision speed, disaster-response utility, area-preservation error, distance error, and user-specific task performance.

For a state-space implementation, one can estimate

$$X_{t+1} = F(X_t, U_t) + \omega_t, \quad (20)$$

$$Y_t = H(X_t) + \varepsilon_t, \quad (21)$$

and maintain posterior uncertainty rather than presenting a map as perfectly known [7].

The data architecture is therefore:

$$\text{Sensors} \rightarrow \text{raw observations} \rightarrow \text{calibration} \rightarrow \text{state estimation} \rightarrow \text{projection} \rightarrow \text{map} \rightarrow \text{decision}. \quad (22)$$

This makes provenance and uncertainty first-class data objects.

8 Information Theory and Artificial Intelligence

A map is an information-bearing representation. If X is the geographic state and M the map, mutual information

$$I(X; M) = \sum_{x, m} p(x, m) \log \frac{p(x, m)}{p(x)p(m)} \quad (23)$$

provides one formal measure of how much information the representation conveys about the underlying state [8].

A projection that is visually attractive but strategically irrelevant can therefore have lower decision value than a less attractive projection that preserves the information most relevant to the task.

Machine learning can operate at several points in the pipeline: classification, segmentation, anomaly detection, feature extraction, state estimation, and projection selection. Let a learned system choose

$$\theta_t = \pi_\phi(Z_t), \quad (24)$$

where Z_t summarizes task characteristics and estimated state. The learning objective may then be written generically as

$$\min_{\phi} \mathbb{E} \left[\mathcal{L} \left(P_{\pi_\phi(Z_t)}(\hat{X}_t), X_t \right) \right]. \quad (25)$$

Reinforcement learning provides a natural extension when map choice changes later decisions. With state s_t , action a_t , and reward R_t ,

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \beta^k R_{t+k} \mid s_t = s, a_t = a \right], \quad (26)$$

following the standard dynamic-programming interpretation [9, 10].

The appropriate role of AI is therefore not to replace cartography. It is to enlarge the set of feasible state-contingent representations while preserving provenance, uncertainty estimates, and human accountability.

9 Political Science, Diplomacy, and Geopolitics

Technology-sharing is itself a political institution. The relevant object is a repeated interaction among jurisdictions with heterogeneous interests, incomplete information, and different strategic exposures. This is consistent with the SNoG political-institutions framework, which emphasizes institutional diversity, separation of risks, proportional authorization, and resistance to concentrated failure [15].

The nuclear-diplomacy framework similarly treats communication, reassurance, credibility, and bargaining as mechanisms for keeping strategic interaction within a survivable region [16]. The international-relations paper adds a strategic-reflexivity perspective: each participant changes the environment in which other participants make decisions [17].

Technology transfer should therefore be structured as a repeated cooperative game rather than a one-time handover. A simple bilateral utility representation is

$$U_i = B_i(T) - C_i(T) + E_i(T) - R_i(T), \quad (27)$$

where B_i denotes direct benefit, C_i transfer and maintenance cost, E_i spillover value, and R_i strategic or governance risk.

The geopolitical dimension is spatial. Routes, ports, chokepoints, weather, political instability, infrastructure, and logistics all depend on geography. The SNoG commerce framework explicitly treats route risk as a function of distance, chokepoints, conflict, piracy, weather, and institutions [14]. The SNoG port papers likewise model ports as spatial nodes embedded in logistics networks [19, 20].

The cartographic system therefore becomes a strategic information infrastructure:

$$\text{Geography} \rightarrow \text{Observation} \rightarrow \text{Knowledge} \rightarrow \text{Coordination} \rightarrow \text{Diplomatic Choice}. \quad (28)$$

10 Welfare Economics and Warfare Economics

Private and social values can diverge. Let U_P denote private benefit and W social welfare. Then

$$W = U_P + E, \quad (29)$$

where E denotes external effects. Technology transfer may create positive externalities through disaster response, navigation, infrastructure planning, scientific cooperation, and educational capacity. It may also create negative externalities through privacy loss, strategic misuse, or information asymmetries.

Accordingly, the socially optimal transfer rule is

$$\max_T \{W(T) - \lambda R(T) - C(T)\}, \quad (30)$$

subject to applicable legal and institutional constraints. This is aligned with the welfare orientation in the SNoG political and commercial frameworks [15, 14].

Warfare economics enters because geographic information can have dual-use characteristics. The present paper deliberately restricts itself to high-level questions of resilience, navigation, logistics, humanitarian response, infrastructure protection, and strategic risk assessment. It does not provide operational targeting instructions. The economic principle is that the marginal social value of a technology depends on both productive applications and the costs of misuse.

11 Physics, Chemistry, and Biology as Structural Analogies

The physical interpretation is straightforward: the globe is a curved state space and the map is a coordinate representation subject to geometric constraints. Coordinate choice does not alter the underlying object; it changes its representation.

The chemical analogy is a transformation network. The underlying spatial state is analogous to an input mixture; observation, calibration, projection, and visualization form a chain of transformations whose intermediate states can be audited.

The biological analogy is adaptation. A fixed map is analogous to a fixed phenotype under changing environmental conditions, whereas a map-making technology provides a capacity for adaptation. The value of that capacity rises with environmental variation.

These analogies are structural rather than claims that economic or cartographic systems literally obey biological or chemical laws. This methodological caution is consistent with the SNoG political-institutions paper [15].

12 Psychology, Psychiatry, Philosophy, and Religion

Human users do not consume maps as neutral databases. They interpret symbols, colors, scale, framing, salience, and omitted information. A projection can therefore alter cognition even when the underlying geographic state is unchanged. This connects cartography to bounded rationality, framing, loss aversion, and probability perception [13].

A cautious psychiatric perspective is relevant only at the level of cognition and human–system interaction: maps are interpreted by people with finite attention, stress responses, and heterogeneous perceptual capacities. The paper makes no clinical inference about individuals.

Philosophically, the globe–map distinction instantiates a broader epistemic principle:

$$\text{Representation} \neq \text{Object represented.} \quad (31)$$

A map is an instrument of knowledge, not the world itself. Different maps can therefore be mutually consistent while preserving different invariants.

The normative dimension can be expressed through stewardship. Knowledge, infrastructure, and technological power can be treated as resources carrying obligations toward others and toward future users. The SNoG commercial framework explicitly treats commerce as both an economic and moral institution, with trust and obligations embedded in repeated exchange [14].

13 Computer Science and Algorithmic Architecture

The complete system can be implemented as a modular pipeline. A minimal algorithm is:

Algorithm ReciprocalCartographicSystem

Input:

Earth-observation streams Y
Portuguese cartographic technology set Θ
Task specification Z
Economic and risk parameters

Repeat:

acquire Y_t
calibrate and validate observations
estimate geographic state $\hat{X}_t$
characterize uncertainty

```

select projection theta_t using Z and X_hat_t
generate map M_t = P_theta_t(X_hat_t)
evaluate map against task-specific loss
store provenance, parameters, and uncertainty
update learned components
evaluate social benefit, cost, and strategic risk

```

Output:

```

map M_t
projection metadata
uncertainty assessment
provenance record
decision-relevant indicators

```

The system should separate raw data, transformed data, estimated state, visualization, and decision layers. Such modularity makes it possible to audit whether an error arose in sensing, inference, projection, or interpretation.

A secure implementation should additionally use access control, versioning, cryptographic provenance, and independent validation. In a distributed environment, no single visualization should automatically become the authoritative state of the system.

14 A Compact Interdisciplinary Synthesis

Table 1: Interpretation of the cartographic technology-transfer problem across fields.

Field	Core object	Contribution to the framework
Cartography	Globe, projection, map	Selection of preserved properties under unavoidable distortion
Remote sensing	Observation	Measurement of latent terrestrial states
Differential geometry	Curvature and metrics	Proof that global planar representation is not lossless
Economics	Technology and product	Distinction between productive capacity and current output
Finance	Real option	Value of state-contingent future choice
Econometrics	Identification	Testing whether technology access changes outcomes
Statistics	Uncertainty	Estimation and confidence about latent states
Information theory	Mutual information	Information content of representations
Computer science	Pipeline	Reproducible modular implementation
AI/ML	Learned policy	Adaptive projection and representation selection
Political science	Institutions	Governance of technological authority
Diplomacy	Repeated bargaining	Credibility, reassurance, and communication

Continued on next page

Field	Core object	Contribution to the framework
International relations	Strategic interaction	Endogenous responses to new information capabilities
Geopolitics	Spatial networks	Routes, ports, chokepoints, and territorial structure
Welfare economics	Externalities	Social benefit versus private benefit
Warfare economics	Dual use	Strategic risk and defensive resilience
Psychology	Cognition	Framing, attention, and interpretation
Philosophy	Epistemology	Representation is not the represented object
Religion and moral economy	Stewardship	Normative responsibility in technology use
Biology	Adaptation	Value of flexible representation under changing states
Chemistry	Transformation networks	Layered transformations and intermediate validation

The table shows that the interdisciplinary structure is not a collection of unrelated metaphors. The common architecture is

$$\boxed{\text{State} \rightarrow \text{Observation} \rightarrow \text{Representation} \rightarrow \text{Inference} \rightarrow \text{Decision}}. \quad (32)$$

What differs by discipline is the object being observed, the transformation rule, the uncertainty model, and the objective function.

15 Institutional Inductance and the Portuguese Option

The SNoG inductance framework defines admission in terms of positive systemic contribution and requires an entrant to be credible, safe, diplomatically responsible, demographically compatible, and economically non-redundant [18]. The same logic can be applied to a technological partnership.

Let Portugal's contribution be

$$V_{\text{PT}} = V_{\text{cart}} + V_{\text{navigation}} + V_{\text{knowledge}} + V_{\text{institutional}}, \quad (33)$$

and SNoG's contribution be

$$V_{\text{SNoG}} = V_{\text{observation}} + V_{\text{computation}} + V_{\text{state estimation}} + V_{\text{automation}}. \quad (34)$$

A reciprocal technology agreement is attractive when

$$V_{\text{PT}} + V_{\text{SNoG}} + V_{\text{synergy}} > C_{\text{integration}} + C_{\text{governance}} + C_{\text{risk}}. \quad (35)$$

The partnership therefore does not require Portugal to become a passive recipient. Nor does it require SNoG to claim exclusive ownership of cartographic production. The point is to create a larger feasible technology space.

This is the cartographic analogue of the broader SNoG principle that the admission of an additional participant should be evaluated by incremental systemic value rather than by membership count alone [18].

16 Limits

The paper is conceptual and does not establish that a real-world institution called the SNoG presently possesses any specified satellite constellation. That claim is treated here as a premise of the SNoG model and dialogue rather than as an independently verified empirical fact.

Likewise, the option-value theorem is a standard consequence of the inequality between an expectation of a maximum and a maximum of expectations. It does not imply that technology transfer is costless, legally unrestricted, or always welfare-improving. Intellectual property, national security, privacy, export controls, cybersecurity, data sovereignty, and institutional trust can all affect the optimal contract.

Finally, maps remain task-dependent. A representation that is optimal for navigation may be inappropriate for statistical area comparison; a map optimized for one projection property may be unsuitable for another [1, 2]. The correct objective is therefore not to discover the one true map, but to build a system capable of choosing appropriate representations from a well-governed technology set.

17 Conclusion

The fundamental distinction is simple:

$$\begin{array}{l} \text{globe} = \text{object}, \quad \text{map} = \text{representation}, \\ \text{cartography} = \text{technology for producing representations.} \end{array} \quad (36)$$

Remote sensing adds another essential layer:

$$\text{Observation} \rightarrow \text{State estimation} \rightarrow \text{Projection} \rightarrow \text{Map}. \quad (37)$$

The economic implication follows immediately. A fixed map is an output. A cartographic system is an option-generating technology. Under uncertainty, the capacity to choose among representations after observing the state can have positive option value.

For that reason, the proposed Portuguese–SNoG bargain should not be framed as a transfer of maps in one direction. It should be framed as reciprocal technological interoperability:

$$\text{Portuguese cartography} \longleftrightarrow \text{SNoG Earth observation}. \quad (38)$$

Lopo Homem therefore represents, in this conceptual dialogue, the Portuguese side’s accumulated cartographic capability; Rajan supplies a natural financial-economic lens for the value of flexibility and future choice; and the SNoG framework supplies institutional concepts of risk, diplomacy, inductance, and systemic value [5, 6, 15, 16, 18].

The final object of the bargain is not a map.

It is the capability to make the *next* map when the world changes.

References

- [1] U.S. Geological Survey, *Map Projections*, General Information Product 70047422, 1993. <https://www.usgs.gov/publications/map-projections>.
- [2] J. P. Snyder, *Map Projections: A Working Manual*, U.S. Geological Survey Professional Paper 1395, U.S. Government Printing Office, 1987. <https://pubs.usgs.gov/publication/pp1395>.

- [3] A. P. Colvocoresses and J. P. Snyder, *Status of Space Oblique Mercator (SOM) Projection (EC 64 Landsat)*, U.S. Geological Survey Open-File Report 78-682, 1978. <https://pubs.usgs.gov/publication/ofr78682>.
- [4] Wikipedia, *Map Projection*, accessed 2026. https://en.wikipedia.org/wiki/Map_projection.
- [5] Library of Congress, *Nautical Atlas of the World, Circular World Map of the Portuguese Hemisphere and Title Page*, Miller Atlas, 1519. <https://www.loc.gov/item/2021668720/>.
- [6] University of Chicago Booth School of Business, *Raghuram G. Rajan*, faculty profile, accessed 2026. <https://www.chicagobooth.edu/faculty/directory/r/Raghuram-G-Rajan>.
- [7] R. E. Kalman, A New Approach to Linear Filtering and Prediction Problems, *Journal of Basic Engineering*, 82(1), 35–45, 1960.
- [8] C. E. Shannon, A Mathematical Theory of Communication, *Bell System Technical Journal*, 27, 379–423 and 623–656, 1948.
- [9] R. Bellman, *Dynamic Programming*, Princeton University Press, 1957.
- [10] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.
- [11] J. M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, 2010.
- [12] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, 2009.
- [13] D. Kahneman and A. Tversky, Prospect Theory: An Analysis of Decision under Risk, *Econometrica*, 47(2), 263–291, 1979.
- [14] S. Ghosh with ChatGPT (OpenAI), *The Real Theory of Commerce: Exchange, Persistence, Loss, Insurance, and Optimal Stopping*, 2026.
- [15] S. Ghosh with ChatGPT (OpenAI), *Political Institutions of the Standard Nuclear oligARCHy*, 2026.
- [16] S. Ghosh with ChatGPT (OpenAI), *Nuclear Diplomacy of the Standard Nuclear oligARCHy: Bargaining, Deterrence, Reassurance, Networks, Catastrophic Risk and Institutional Survival*, 2026.
- [17] S. Ghosh with ChatGPT (OpenAI), *The International Relations Theory of the Standard Nuclear oligARCHy*, 2026.
- [18] S. Ghosh, *On the Inductance of Deserving Nations into the Standard Nuclear oligARCHy*, 2026.
- [19] S. Ghosh with ChatGPT (OpenAI), *Logistic Chain Equilibrium on Spherical Port Networks: An Entropic Optimal Transport Approach*, 2026.
- [20] S. Ghosh with ChatGPT (OpenAI), *A State-of-the-Art Theory of the Containerized Port: Queuing, Security, Classification, Multi-Currency Settlement, Inventory Risk Management, and Hinterland Logistics*, 2026.

Glossary

Cartography

The science, technology, and practice of representing geographic information.

Globe

A three-dimensional representation of the terrestrial surface that preserves the underlying curved geometry more directly than a planar map.

Map

A representation of geographic information in a chosen coordinate and visual system.

Map projection

A mathematical transformation that represents part or all of a curved surface on a plane.

Projection distortion

The geometric discrepancy introduced when a curved surface is represented on a plane.

Earth observation

Measurement of terrestrial states using remote sensors, including satellite-based systems.

State estimation

Statistical inference about an unobserved or partially observed system state.

Technology transfer

Transfer of a productive capability that enables generation or adaptation of future outputs.

Product transfer

Transfer of a particular completed output, such as a fixed map.

Option value

The economic value of retaining the right, without an obligation, to choose among future actions after uncertainty resolves.

Inductance

In the SNoG framework, the systemic capacity to admit an external participant when the participant contributes sufficient positive value subject to stated constraints.

Systemic value

Incremental value generated for an interconnected system as a whole rather than for a single participant alone.

Provenance

A record of the sources, transformations, parameters, and processing steps used to produce a result.

Dual use

A property of technology that permits both beneficial civilian applications and applications with strategic or security implications.

Survivable region

A region of the state space in which catastrophic-risk constraints remain satisfied in the SNoG diplomatic framework.

The End